

CSA0614- TOPIC 3-1 -DAA

Y Greeshma Reddy

192424146

SIMATS Online Lab

CSA0614-DAA/CO4 ASSESSME

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SIMATS Online Programming Lab

YAMMALA GREESHMA REDDY

LAB PROGRAM - TOPIC 3-1

Language: Python C++ Java

QUESTIONS

- Q1 Sequential Search 1 mark
- Q2 Sequential Search 1 mark
- Q3 Sequential Search 1 mark
- Q4 Sequential Search 1 mark
- Q5 Sequential Search 1 mark
- Q6 Sequential Search 1 mark
- Q7 Sequential Search 1 mark
- Q8 Sequential Search 1 mark
- Q9 Sequential Search 1 mark
- Q10 Sequential Search 1 mark

10/10 answered

Sequential Search 1 mark

Implement Sequential Search and analyze its performance.

Input:

Array:
45 22 87 12 88 34 96 78 90 11
28 75 18 64 37

Search Key:
75

18
100

Tasks:

- Show each comparison.
- Count comparisons for every search.
- Identify best, average, and worst cases.
- Complexity
- Best-case complexity
- Average-case complexity
- Worst-case complexity
- Space complexity

Your Code

```
def sequential_search(arr, key):
    for i in range(len(arr)):
        comparison += 1
        if arr[i] == key:
            return i, comparison
    return -1, comparison

arr = [45, 22, 87, 12, 88, 34, 96, 78, 90, 11, 28, 75, 18, 64, 37]
key = 75

search_key = [75, 18, 100]

for key in search_key:
    index, comparison = sequential_search(arr, key)
    print(f"Searching for key: {key}")
    if index != -1:
        print(f"Comparison made: {comparison}")
        print(f"Element found at index: {index}")
    else:
        print(f"Element not found")
```

Input:

75
18
100

Output:

Searching for key: 75
Comparison made: 12
Element found at index: 11
Searching for key: 18
Comparison made: 10
Element found at index: 13
Searching for key: 100
Comparison made: 15
Element not found

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- Q10 Sequential Search 1 mark

10/10 answered

Sequential Search 1 mark

Write a program to perform Sequential Search on a 2-D matrix.

Input:

12 8 15
5 18 27
9 11 24

Key = 28

Output:

Element found at Row 3 Column 3

Your Code

```
def sequential_search_2d(matrix, key):
    rows = len(matrix)
    cols = len(matrix[0])

    for i in range(rows):
        for j in range(cols):
            if matrix[i][j] == key:
                return i, j, "Element found at Row {} Column {}".format(i+1, j+1)

    return -1, -1, "Element not found"

matrix = [
    [12, 8, 15],
    [5, 18, 27],
    [9, 11, 24]
]
key = 28

row, col, message = sequential_search_2d(matrix, key)
print(f"Searching for key: {key}")
if row != -1:
    print(f"Comparison made: {message}")
else:
    print(f"Element not found")
```

Input:

12 8 15
5 18 27
9 11 24

Output:

Element found at Row 3 Column 3

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MANJALA GRESHMA REDDY

LAB PROGRAM- TOPIC 3-1

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Q8 Sequential Search (1 mark)

Q9 Sequential Search (1 mark)

Q10 Sequential Search (1 mark)

10/10 answered

Sequential Search

1 mark

Implement Sequential Search for string data.

Input:

Name: Aha, Bala, Charan, Deepa, Esha, Farhan

Search Name: Deepa

Output: Name found at position 4

Your Code

```
def sequential_search(name, search_name):  
    for i in range(len(name)):  
        if name[i] == search_name:  
            return i + 1 # position (1-based indexing)  
    return -1  
  
# Input  
name = input()  
search = input()  
# Search  
position = sequential_search(name, search)  
  
# Output  
if position != -1:  
    print("Name found at position", position)  
else:  
    print("Name not found")
```

Input

Aha, Bala, Charan, Deepa, Esha, Farhan

Deepa

Output

Name found at position 4

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Q9 Sequential Search (1 mark)

Q10 Sequential Search (1 mark)

10/10 answered

Sequential Search

1 mark

Write a program to perform Sequential Search on an array of student register numbers.

Input:

Register Numbers: 101, 102, 103, 104, 105, 106

Search Key: 103

Output: Register Number found at position 4

Your Code

```
def sequential_search(arr, key):  
    for i in range(len(arr)):  
        if arr[i] == key:  
            return i + 1  
    return -1  
  
# Input  
arr = [101, 102, 103, 104, 105, 106]  
key = 103  
  
# Search  
pos = sequential_search(arr, key)  
  
# Output  
if pos != -1:  
    print("Register Number found at position", pos)  
else:  
    print("Register Number not found")
```

Input

101, 102, 103, 104, 105, 106

103

Output

Register Number found at position 4

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Q6 Sequential Search 1 mark

Q7 Sequential Search 1 mark

Q8 Sequential Search 1 mark

Q9 Sequential Search 1 mark

Q10 Sequential Search 1 mark

10/10 answered

Sequential Search 1 mark

Implement Sentinel Sequential Search and compare it with ordinary Sequential Search.

Input
Array: 14, 8, 22, 35, 16, 41, 27
Key: 15

Output
Position found
Comparison count:

Your Code

```
def sequential_search(arr, key):  
    comparisons = 0  
    for i in range(len(arr)):  
        comparisons += 1  
        if arr[i] == key:  
            return i, comparisons  
    return -1, comparisons  
  
def sentinel_sequential_search(arr, key):  
    # Sentinel Sequential Search: Modifies a copy  
    # of arr[] to copy the original array element  
    # at the end  
    comparisons = 0  
    last = arr[-1]  
    arr[-1] = key # put sentinel at last position  
    i = 0  
    while True:  
        comparisons += 1  
        if arr[i] == key:  
            break  
        i += 1  
    arr[-1] = last # remove original last element  
    # If found at last position, check if it was truly found or sentinel  
    if i < len(arr) - 1 or last == key:  
        return i, comparisons  
    return -1, comparisons  
  
arr = [14, 8, 22, 35, 16, 41, 27]  
key = 15  
pos, cnt = sequential_search(arr, key)  
pos2, cnt2 = sentinel_sequential_search(arr, key)  
print("Normal Sequential Search:")  
print("Position found", pos, cnt)
```

Input
Array: 14, 8, 22, 35, 16, 41, 27
Key: 15

Output
Position found: 4
Comparison count: 8
Sentinel Sequential Search:

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Q6 Sequential Search 1 mark

Q7 Sequential Search 1 mark

Q8 Sequential Search 1 mark

Q9 Sequential Search 1 mark

Q10 Sequential Search 1 mark

10/10 answered

Brute Force String Matching 1 mark

Write a complete program for Brute Force String Matching and perform complexity analysis.

Input
Text = "THIRUGARICURUTOTTHIRUGARICURUTOT"
Pattern = "TARUT"

Tasks
- Find all occurrences of the pattern.
- Count total comparisons.
- Display the shifting process.
- Analyze:
- Best-case complexity
- Worst-case complexity
- Space complexity

Your Code

```
def brute_force_string_matching(text, pattern):  
    n = len(text)  
    m = len(pattern)  
    comparisons = 0  
    total_comparisons = 0  
    for i in range(n - m + 1):  
        print("Text:", text)  
        print("Pattern:", pattern)  
        print("Matching process:", i)  
        # Check if the starting index of the pattern appears in text  
        for j in range(m):  
            print("Match at index", i, ":", text[i+j] == pattern[j])  
            comparisons += 1  
            total_comparisons += 1  
        if comparisons == m:  
            print("Compare pattern at index", i, "with Text")  
            if pattern[i:i+m] == text[i:i+m]:  
                print("Match found at index", i)  
            else:  
                print("No match found at index", i)  
            comparisons = 0  
        else:  
            print("Pattern does not match at index", i)  
            comparisons = 0  
    print("Total Comparisons:", total_comparisons)
```

Input
Text: THIRUGARICURUTOTTHIRUGARICURUTOT
Pattern: TARUT

Output
Match found at index 10
Match found at index 20

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DATA PROGRAMMING

MINIMALA GRESHMA REDDY

LAB PROGRAM - TOPIC 3-1

Language Python C++ Java

QUESTIONS

Q1 Requested Search 1 mark

Q2 Requested Search 1 mark

Q3 Requested Search 1 mark

Q4 Requested Search 1 mark

Q5 Requested Search 1 mark

Q6 Requested Search 1 mark

Q7 Brute Force String Matching 1 mark

Q8 Brute Force String Matching 1 mark

Q9 Brute Force String Matching 1 mark

Q10 Brute Force String Matching 1 mark

10/10 answered

Brute Force String Matching 1 mark

Implement Brute Force String Matching and display every alignment of the pattern with the text.

Input:

- Text = "ABCDCABACDCA"
- Pattern = "ACDCA"

Output:

- Alignment number
- Matching result
- Pattern occurrence positions

Your Code

```
def brute_force_string_matching(text, pattern):
    n = len(text)
    m = len(pattern)
    matches = []

    # Iterate through all possible alignments
    for i in range(n - m + 1):
        match = True

        # Check if pattern matches text at current alignment
        for j in range(m):
            if text[i + j] != pattern[j]:
                match = False
                break

        if match:
            matches.append(i)
            print(f"Alignment ({i}): Match found at position ({i})")
        else:
            print(f"Alignment ({i}): No match")

    return matches

# Example usage
text = "ABCDCABACDCA"
pattern = "ACDCA"

# Function to display results
def display_results(text, pattern, matches):
    print(f"Brute Force String Matching Results ----")
    print(f"Pattern = '{pattern}'")
    print(f"Text = '{text}'")
    print(f"Pattern occurrence positions: {matches}")

# Call the function
brute_force_string_matching(text, pattern)
display_results(text, pattern, brute_force_string_matching(text, pattern))
```

Input

Enter Input

Output

Brute Force String Matching Results ----
Pattern = ACDCA
Text = ABCDCABACDCA
Alignment 1: Match found at position 1
Alignment 2: No match
Alignment 3: No match

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Q5 Requested Search 1 mark

Q6 Requested Search 1 mark

Q7 Brute Force String Matching 1 mark

Q8 Brute Force String Matching 1 mark

Q9 Brute Force String Matching 1 mark

Q10 Brute Force String Matching 1 mark

10/10 answered

Brute Force String Matching 1 mark

Write a program to compare the execution of Brute Force String Matching for:

- Successful search
- Unsuccessful search

Input:

- Text = "ACDCAHUNGSLAP"
- Pattern 1 = "SLAP"
- Pattern 2 = "TEST"

Display the comparison count for both cases.

Your Code

```
def brute_force_string_matching(text, pattern):
    n = len(text)
    m = len(pattern)
    matches = 0
    mismatches = 0

    for i in range(n - m + 1):
        match = True

        for j in range(m):
            if text[i + j] == pattern[j]:
                matches += 1
            else:
                match = False
                mismatches += 1
                break

        if match:
            print(f"Successful match at position {i}")
        else:
            print(f"Unsuccessful match at position {i}")

    return matches, mismatches

def main():
    text = "ACDCAHUNGSLAP"
    pattern1 = "SLAP"
    pattern2 = "TEST"

    # Successful search (Pattern 1 is in the text)
    matches1, mismatches1 = brute_force_string_matching(text, pattern1)
    print(f"Successful match (Pattern 1) at position {matches1}")

    # Unsuccessful search (Pattern 2 is not in the text)
    matches2, mismatches2 = brute_force_string_matching(text, pattern2)
    print(f"Unsuccessful match (Pattern 2) at position {mismatches2}")

    print(f"Total Successful Matches: {matches1}")
    print(f"Total Unsuccessful Matches: {mismatches2}")

    if __name__ == "__main__":
        main()
```

Input

Enter Input

Output

Successful match (Pattern 1) at position 10
Unsuccessful match (Pattern 2) at position 14
Total Successful Matches: 1
Total Unsuccessful Matches: 14

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Q3 Required Search 1 mark

Q4 Required Search 1 mark

Q5 Required Search 1 mark

Q6 Brute Force String Matching 1 mark

Q7 Brute Force String Matching 1 mark

Q8 Brute Force String Matching 1 mark

Q9 Brute Force String Matching 1 mark

Q10 Brute Force String Matching 1 mark

10/10 answered

Brute Force String Matching 1 mark

Implement Brute Force Matching for case-insensitive searching.

Input:

- Text = "DataStructuresAndAlgorithms"
- Pattern = "SUBSTRINGS"

Output:

- Pattern found position

Your Code

```
def brute_force_string_matching(text, pattern):
    n = len(text)
    m = len(pattern)
    pattern_index = pattern.index()
    i = 0
    j = 0
    k = 0
    l = 0
    m = len(pattern)
    n = len(text)
    for i in range(n - m + 1):
        for j in range(m):
            if text[i+j] == pattern[j]:
                continue
            else:
                break
        if j == m - 1:
            return i
    return -1
```

Input

Enter Input

Output

Submit

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Q7 Brute Force String Matching 1 mark

Q8 Brute Force String Matching 1 mark

Q9 Brute Force String Matching 1 mark

Q10 Brute Force String Matching 1 mark

10/10 answered

Brute Force String Matching 1 mark

Write a program that searches for a pattern in a text and reports the first occurrence only using Brute Force String Matching.

Input:

- Text = "COMPUTERSCIENCE"
- Pattern = "SC"

Output:

- First occurrence position
- Number of comparisons

Your Code

```
def brute_force_string_matching(text, pattern):
    n = len(text)
    m = len(pattern)
    found_index = -1
    comparisons = 0
    for i in range(n - m + 1):
        for j in range(m):
            if text[i+j] == pattern[j]:
                continue
            else:
                break
        if j == m - 1:
            found_index = i
            comparisons += 1
    return found_index, comparisons
```

Input

Enter Input

Output

First occurrence position: 5
Number of comparisons: 11

Submit

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